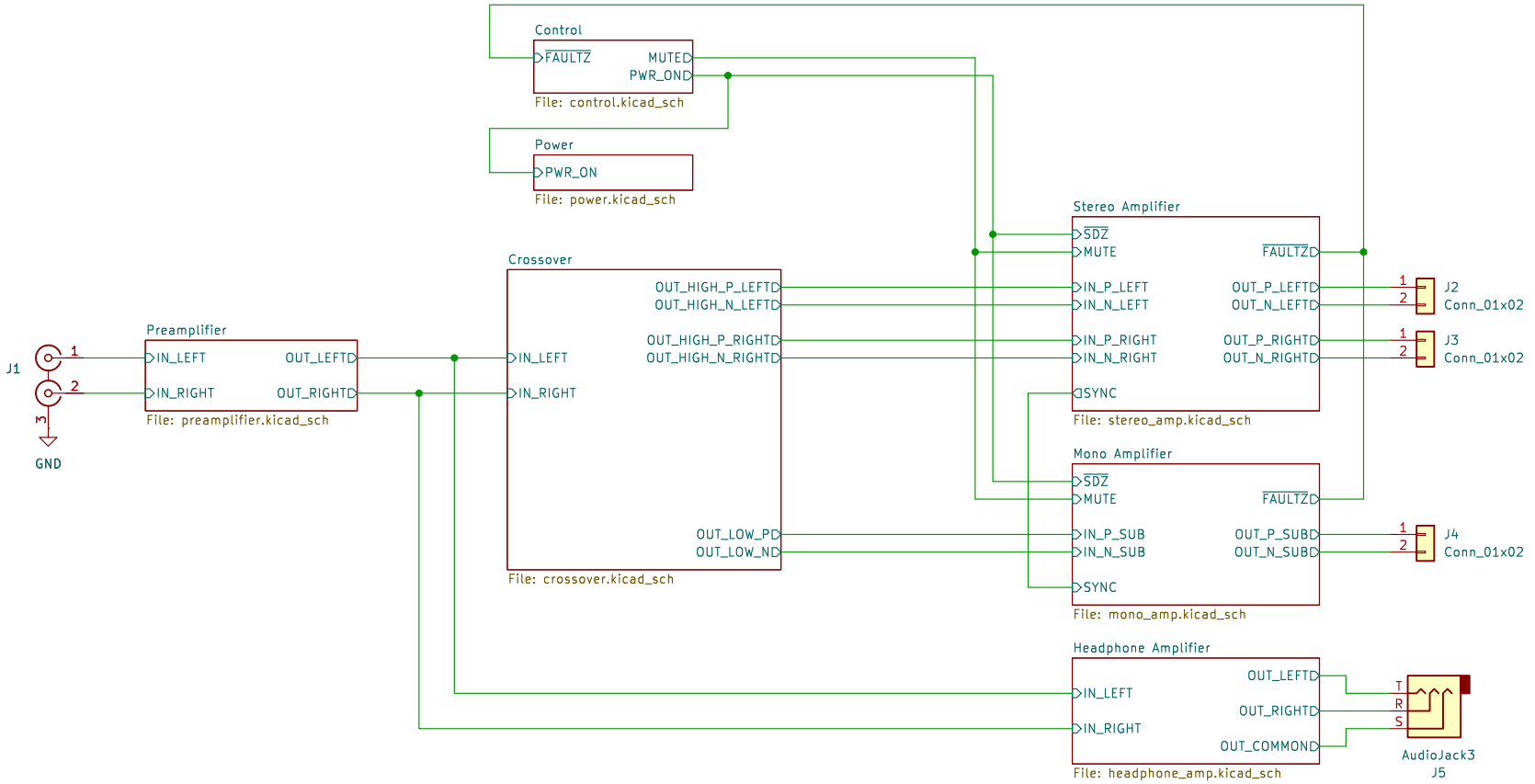
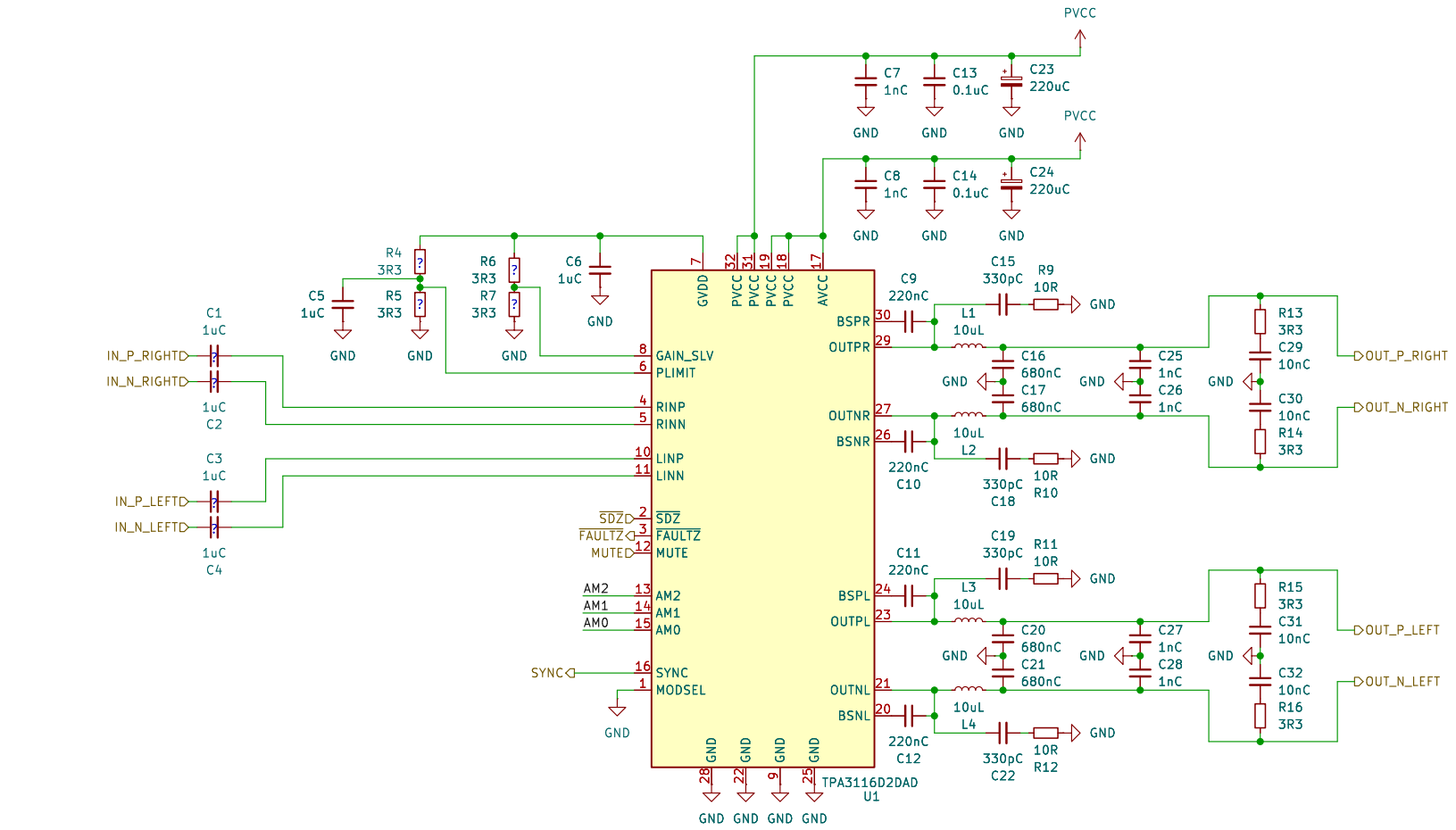
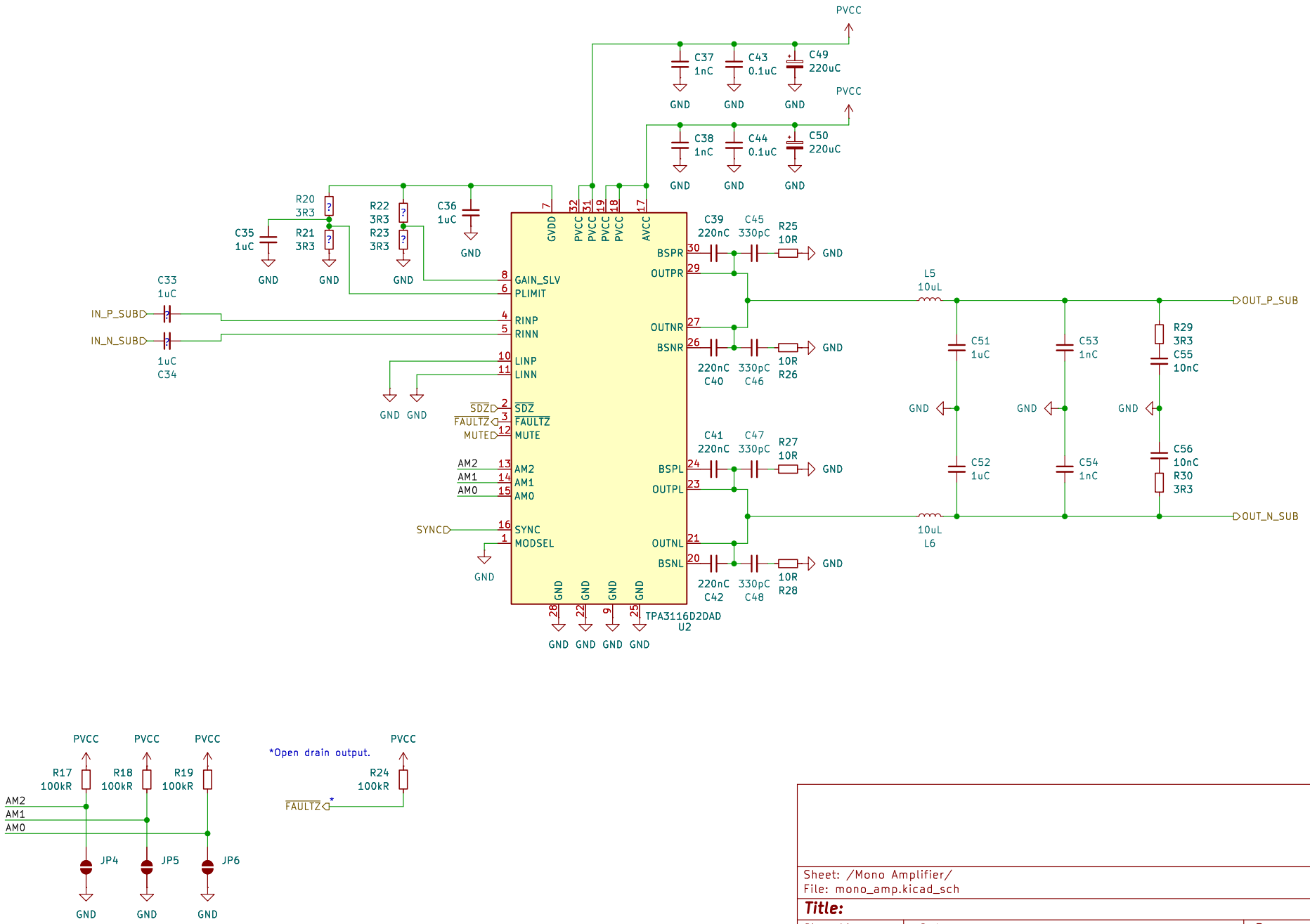


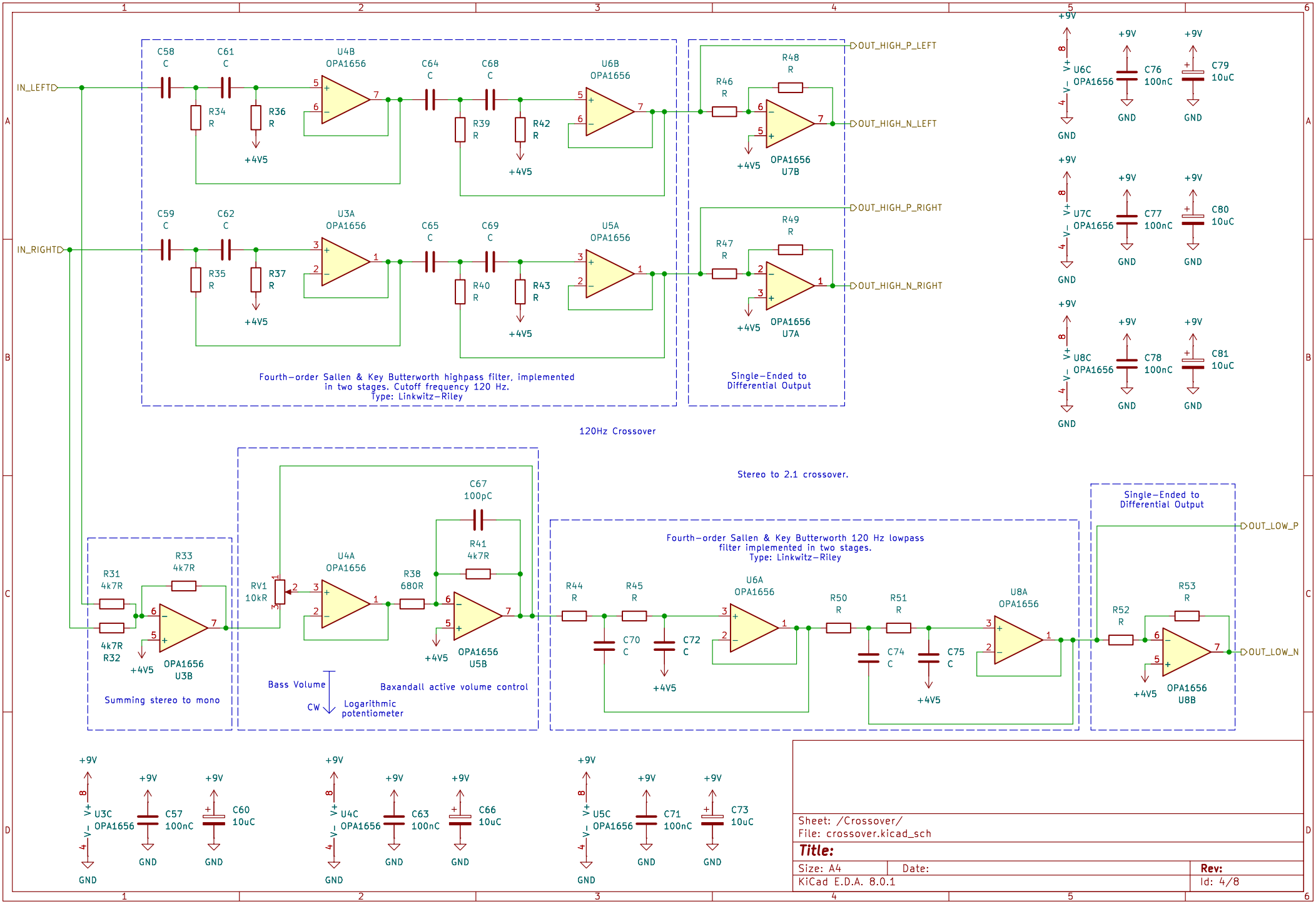
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 1/8

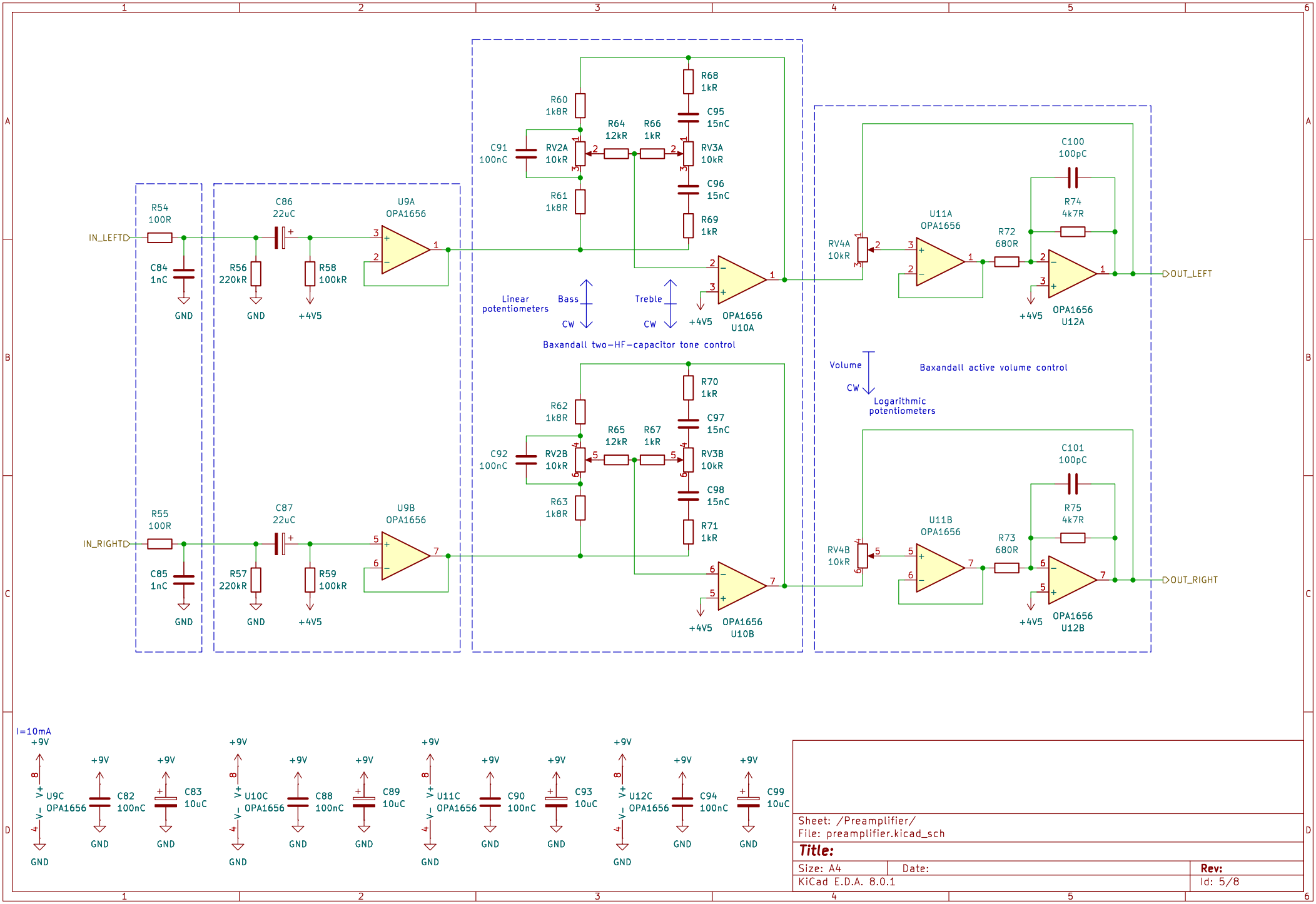


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Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 2/8



Sheet: /Mono Amplifier/		
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Title:		
Size: A4	Date:	**Rev:**
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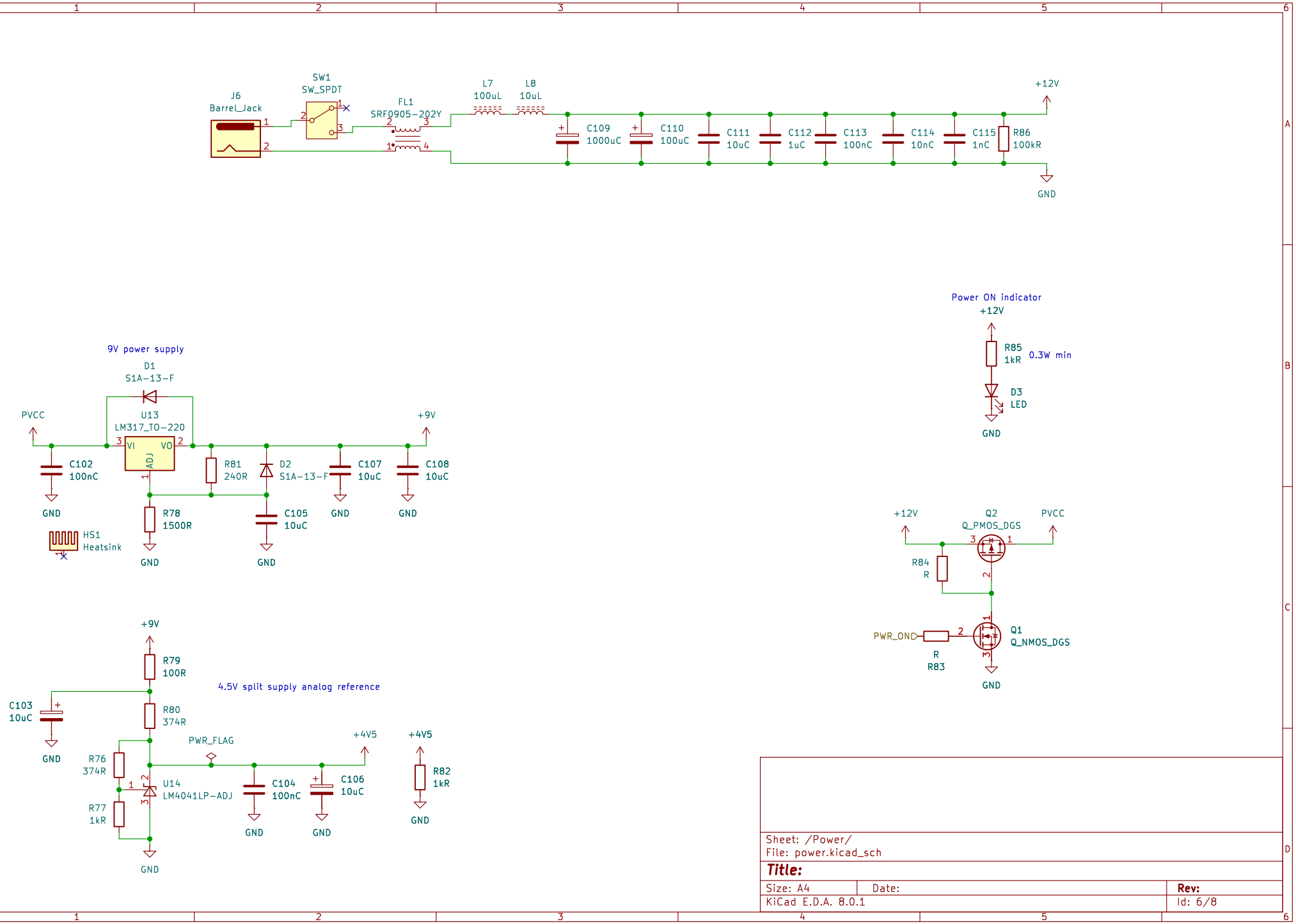

IN_LEFT
IN_RIGHT
C58 C
C61 C
C59 C
C62 C
U4B OPA1656
U3A OPA1656
U6B OPA1656
U5A OPA1656
C64 C
C68 C
C65 C
C69 C
R34 R
R36 R
R35 R
R37 R
R39 R
R42 R
R40 R
R43 R
+4V5
Fourth-order Sallen & Key Butterworth highpass filter, implemented in two stages. Cutoff frequency 120 Hz.
Type: Linkwitz-Riley
R46 R
R48 R
R47 R
R49 R
OPA1656 U7B
OPA1656 U7A
OUT_HIGH_P_LEFT
OUT_HIGH_N_LEFT
OUT_HIGH_P_RIGHT
OUT_HIGH_N_RIGHT
Single-Ended to Differential Output
120Hz Crossover
+9V
GND
U6C OPA1656
U7C OPA1656
U8C OPA1656
C76 100nC
C77 100nC
C78 100nC
C79 10uC
C80 10uC
C81 10uC
Stereo to 2.1 crossover.
R31 4k7R
R33 4k7R
4k7R R32
OPA1656 U3B
Summing stereo to mono
RV1 10kR
U4A OPA1656
R38 680R
C67 100pC
R41 4k7R
OPA1656 U5B
Bass Volume
CW
Logarithmic potentiometer
Baxandall active volume control
Fourth-order Sallen & Key Butterworth 120 Hz lowpass filter implemented in two stages.
Type: Linkwitz-Riley
R44 R
R45 R
U6A OPA1656
R50 R
R51 R
U8A OPA1656
C70 C
C72 C
C74 C
C75 C
Single-Ended to Differential Output
R52 R
R53 R
OPA1656 U8B
OUT_LOW_P
OUT_LOW_N
U3C OPA1656
U4C OPA1656
U5C OPA1656
C57 100nC
C63 100nC
C71 100nC
C60 10uC
C66 10uC
C73 10uC
Sheet: /Crossover/
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Title:
Size: A4
Date:
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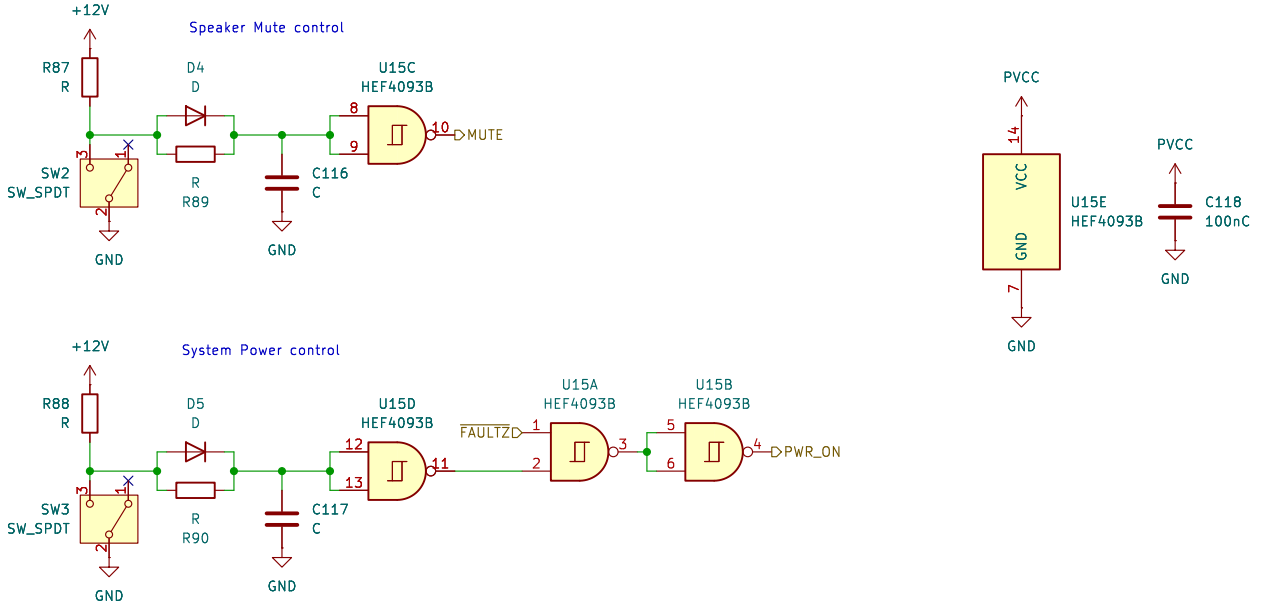


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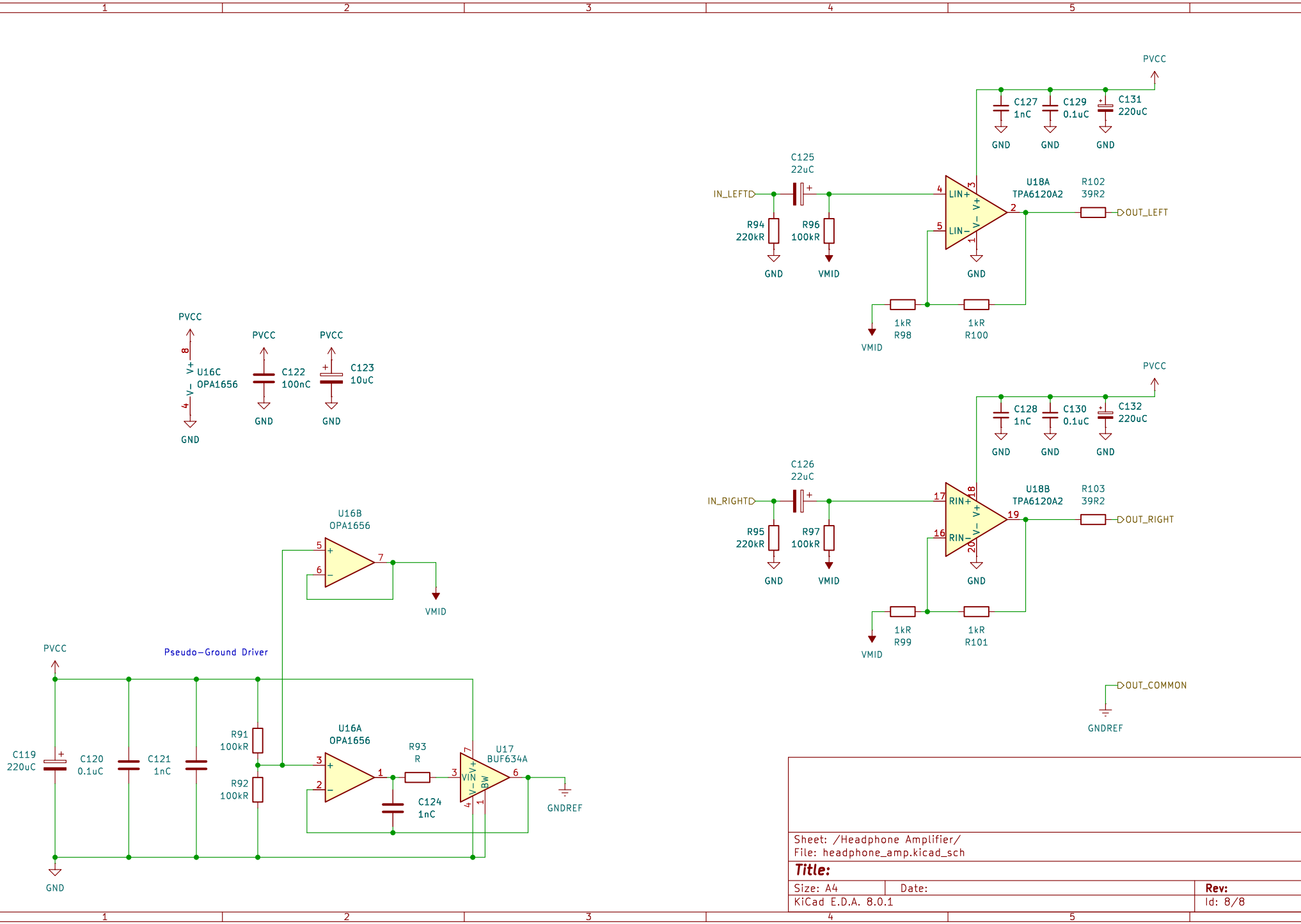
Title:

Size: A4 | Date: | **Rev:**
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J6
Barrel_Jack
SW1
SW_SPDT
FL1
SRF0905-202Y
L7
100uL
L8
10uL
+12V
C109
1000uC
C110
100uC
C111
10uC
C112
1uC
C113
100nC
C114
10nC
C115
1nC
R86
100kR
GND
Power ON indicator
+12V
R85
1kR
0.3W min
D3
LED
GND
9V power supply
D1
S1A-13-F
PVCC
U13
LM317_TO-220
VI
VO
ADJ
+9V
C102
100nC
R81
240R
D2
S1A-13-F
C107
10uC
C108
10uC
R78
1500R
C105
10uC
HS1
Heatsink
GND
+12V
Q2
Q_PMOS_DGS
PVCC
R84
R
PWR_ON
R
R83
Q1
Q_NMOS_DGS
+9V
R79
100R
4.5V split supply analog reference
C103
10uC
R80
374R
PWR_FLAG
+4V5
+4V5
R76
374R
U14
LM4041LP-ADJ
C104
100nC
C106
10uC
R82
1kR
R77
1kR
Sheet: /Power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
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Sheet: /Control/		
File: control.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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PVCC
C127
1nC
C129
0.1uC
C131
220uC
GND
GND
GND
C125
22uC
IN_LEFT
U18A
TPA6120A2
R102
39R2
OUT_LEFT
R94
220kR
R96
100kR
GND
VMID
GND
1kR
R98
1kR
R100
VMID
PVCC
C128
1nC
C130
0.1uC
C132
220uC
GND
GND
GND
C126
22uC
IN_RIGHT
U18B
TPA6120A2
R103
39R2
OUT_RIGHT
R95
220kR
R97
100kR
GND
VMID
GND
1kR
R99
1kR
R101
VMID
OUT_COMMON
GNDREF
PVCC
U16C
OPA1656
GND
PVCC
C122
100nC
GND
PVCC
C123
10uC
GND
U16B
OPA1656
VMID
PVCC
Pseudo-Ground Driver
C119
220uC
C120
0.1uC
C121
1nC
R91
100kR
R92
100kR
U16A
OPA1656
R93
R
C124
1nC
U17
BUF634A
GNDREF
GND
Sheet: /Headphone Amplifier/
File: headphone_amp.kicad_sch
Title:
Size: A4
Date:
Rev:
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